

Week 14 — 🎉 Big Simulations

Name: ____ Date: ____

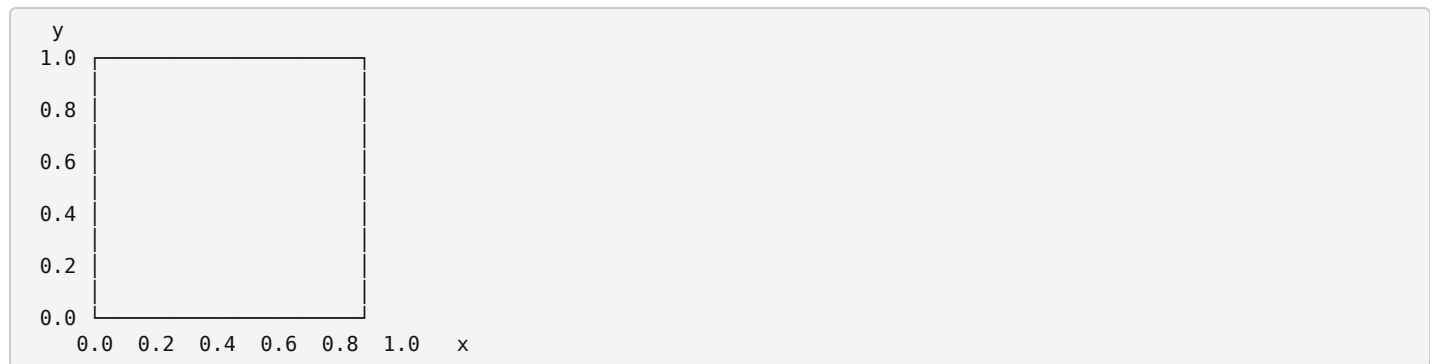
Monte Carlo method: answer a hard question with a mountain of random tries. Fraction of raindrops inside = fraction of area inside. Rain can measure.

Today's words

Word	What it means
Monte Carlo method	answering a hard question with tons of random tries
rand	a random <i>decimal</i> between 0 and 1 — any decimal at all
trial	one single random try (one raindrop, one game)
expected value	your average result <i>per game</i> : each outcome × its probability, added up

1 • Be the raindrop 🌧️

The quarter circle's arc runs from (0, 1) to (1, 0) — every arc point is exactly 1 from the corner (0, 0). Sketch the arc on the grid, plot the five drops, then **compute**: inside means $x^2 + y^2 \leq 1$.



Drop	x	y	x^2	y^2	$x^2 + y^2$	≤ 1 ? in or out?
A	0.3	0.4				
B	0.6	0.8				
C	0.9	0.9				
D	0.5	0.5				
E	0.7	0.8				

2 • Audit the carnival 🎪

Lucky Die: pay \$4, roll one die, win \$1 per dot. Fill in the table, then the verdict.

Roll	Payout	Probability	Payout × probability
1	\$1	1/6	\$
2	\$2	1/6	\$
3	\$3	1/6	\$
4	\$4	1/6	\$
5	\$5	1/6	\$
6	\$6	1/6	\$

Expected payout (add the last column): \$__

Per game at \$4 a ticket, a player averages: \$__ · After 1,000 games, the booth is up about: \$__

🧠 Brain teaser (optional — take it home)

Design a sneaky carnival game. Invent a game (coins, dice, spinners — your call) that *looks* totally fair but has a **negative expected value** for the player. Write:

1. The rules and the ticket price (make the sign tempting!)
2. The expected-value math that exposes the trick

Bring it next week — **the sneakiest game gets simulated in class**, 100,000 plays, so the numbers better back you up.